

Syed Mustafa Mohiuddin

Mechatronics Systems Engineer — Robotics, Motion Control & Embedded Systems

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PROFESSIONAL SUMMARY

Mechatronics and systems engineer with 3+ years of experience in safety-critical motion control, autonomous robotics, and embedded systems. Skilled in real-time control, sensor integration, and regulatory-compliant development at Siemens Healthineers and Kraken Robotics. Experienced across mechanical CAD, electrical design, and embedded software, with a strong focus on automation, robotics, and advanced control systems.

ENGINEERING EXPERIENCE

Siemens Healthineers AG, Kempten

Working Student – Systems Engineering

March 2025 – Present

- Developed and validated a multi-model force limitation algorithm spanning all CT motion system variants, enabling unified single-click activation without hardware modification.
- Performed advanced diagnostics and parameter tuning across drive systems using Drive Manager and TCB, replacing manual routines with scripted, repeatable test sequences.
- Led cross-functional PLM lifecycle tasks in collaboration with mechanical, electrical, and software sub-teams within an international engineering environment.
- Conducted risk evaluation and safety compliance assessments for regulated dynamic motion systems, generating evidence packages aligned with IEC and ISO standards.

Siemens Healthineers AG, Kempten

Mechatronics Intern – Collision Avoidance & LiDAR Integration

April 2024 – February 2025

- Integrated SICK LiDAR sensors into the Nexaris Angio CT sliding gantry architecture, delivering real-time spatial awareness with embedded C/C++ detection logic and safety-rated PLC braking.
- Eliminated false-positive collision triggers through signal filtering and adaptive threshold calibration, removing the need for manual operator intervention.
- Designed circuit diagrams, cable routing layouts, and mechanical sensor mounts from concept to validated installation.
- Maintained full PLM lifecycle traceability and authored compliance documentation throughout the development cycle.

Kraken Robotics, Rostock

Electrical Engineering Intern – Autonomous Underwater Vehicles

October 2023 – March 2024

- Led MATLAB-driven battery discharge analysis for AUVs operating in harsh marine environments, modelling discharge curves against depth, temperature, and mission profiles – yielding an 8% improvement in operational uptime.
- Built telemetry dashboards enabling field deployment teams to make mission-critical power management decisions in real time.
- Conducted environmental stress testing of marine battery systems and performed FMEA on subsea power distribution architectures.
- Supported diagnostics and fault isolation on autonomous platforms under operational conditions.

CORE COMPETENCIES

Motion Control & Robotics

Force-controlled motion systems; real-time torque algorithms; PID tuning; gantry and multi-axis systems; autonomous navigation; ROS/ROS2 integration.

Embedded Systems & Software

C/C++ embedded programming; STM32 microcontrollers; MATLAB/Simulink modelling and HIL validation; real-time control; sensor fusion; AWS IoT pipelines.

Electrical & Sensor Design

Circuit diagram creation; wiring harness design; SICK LiDAR integration; motor drive systems; battery system modelling; EMC and insulation compliance.

Mechanical & CAD Design

Siemens NX; SolidWorks; Creo Parametric; tolerance stack-up analysis; DFM/DFA; sheet metal design; 3D printing and rapid prototyping.

Systems Engineering & Safety

FMEA/FMECA; risk assessment; HIL testing; digital twin modelling; PLM lifecycle management; telemetry analysis; ISO 15066; IEC 60601-2-44; IEC 61508.

TECHNICAL SKILLS

Control & Simulation

MATLAB/Simulink, HIL Testing, Digital Twin, PID Control, Real-Time Control, Drive Manager, TCB

Programming

C/C++, Python, MATLAB, STM32/Arduino, ROS/ROS2, Embedded C, AWS IoT

Aqua Tech GmbH, Germany

Mechanical Design & Production Engineer

March 2024 – August 2024

- Managed end-to-end production control for complex electromechanical assemblies, from design release to floor-level quality verification.
- Produced Siemens NX CAD models with integrated DFM constraints and tolerance stack-up analysis to ensure repeatability and manufacturability.
- Achieved a 10% throughput bottleneck reduction through WIP simulation and production line redesign.
- Designed custom jigs, fixtures, and tooling to support assembly consistency and quality across production runs.

Siemens Mobility, Germany

Operations Industrial Engineer

2022 – 2023

- Supported industrial operations engineering across the rail systems manufacturing division, contributing to process optimisation and production line analysis.
- Coordinated cross-team activities in a high-complexity manufacturing environment and completed the Siemens Mobility Operations Industrial Engineer certification programme.

EDUCATION

Rhine-Waal University of Applied Sciences, Kleve — *B.Eng. Mechatronics Systems Engineering*

February 2026

Control Systems, Automation, Sensors & Actuators, Robotics, Embedded Systems

Rhine-Waal University of Applied Sciences, Kleve — *M.Sc. Engineering in Bionics (Ongoing)*

Expected 2027

Adaptive systems, control theory, bio-inspired robotics, sustainable engineering

Cambridge International Examinations, Karachi — *A-Levels*

2020

KEY ENGINEERING PROJECTS

Intelligent Force Limitation – Medical CT Motion — *Siemens Healthineers, Bachelor Thesis*

Developed a real-time torque-limited force control system for CT gantry motion, enabling unified one-click activation across hardware variants and supporting IEC 60601-2-44 / ISO 15066 compliance validation.

LiDAR-Guided Collision Avoidance System — *Siemens Healthineers*

Implemented a LiDAR-based collision avoidance system for Nexaris Angio CT gantry motion using SICK sensors, embedded C/C++, and PLC braking logic, achieving validated false-positive-free operation.

AUV Power Optimisation & Telemetry — *Kraken Robotics*

Optimized AUV power usage through MATLAB battery modelling and telemetry analytics, improving operational uptime by 8%.

Digital Twin – Precision Motion Dynamics — *Siemens Healthineers*

Built a MATLAB/Simulink digital twin for CT gantry motion dynamics, reducing physical control-loop tuning iterations by over 60%.

Autonomous Robotics & Embedded Control — *University / Independent*

Developed autonomous robotic systems using ROS2, sensor fusion, and STM32-based real-time motor control for embedded navigation and decision-making.

CAD & Mechanical

Siemens NX, SolidWorks, Creo Parametric, Fusion 360, DFM/DFA, Tolerance Analysis, Sheet Metal Design, FEA

Electrical & Sensing

SICK LiDAR, PLC Programming (Safety), PCB Design, Sensor Integration, Motor Control, Power Electronics, Battery Systems, EMC

Standards & Methods

ISO 15066, IEC 60601-2-44, IEC 61508, FMEA/FMECA, PLM, Risk Assessment, Regulatory Documentation

CERTIFICATIONS & AWARDS

Winner – Innovation Think Tank (ITT)

Siemens Healthineers – received acquisition offer for innovative engineering solution.

Best Delegation – Harvard Model UN (HNMUN 2022)

Harvard University, Cambridge, MA.

Siemens NX – Professional CAD Certification

AWS IoT – Microcontroller Integration

Google Analytics – Data & Performance

Energy Resource Systems – Technical Frameworks

Siemens Mobility – Operations Industrial Engineer

R&D Program – Advanced Innovation Methodologies

LANGUAGES

English C1 – Fluent (written & spoken)

German B1 – Conversational & technical

Urdu / Hindi C2 – Native

OTHER

Team Lead – Rhein-Waal Racing Team

Led cross-disciplinary student engineering team through full design, build, and competition cycle for university racing events.

Competitive Judoka

Former professional-level judoka; discipline and precision applied to engineering problem-solving.

Available: Immediately

Location: Bayreuth, DE – open to relocation

Travel: Open to international travel

REFERENCES

Ebersberger, Lucas

Head of Systems Engineering

Siemens Healthineers AG

Contact details available upon request.